

Antarctic Tundra

The world's most inhospitable continent for vegetation, where only 1 percent is covered in plant life



Given that around 99 percent of Antarctica is covered in ice, it is surprising that any plants survive on Earth's southernmost, most elevated continent. Ice sheets, glaciers, and mountains make up its interior, yet small patches of tundra exist, mainly on the Antarctic Peninsula and several sub-Antarctic islands, as well as on a few exposed inland rocks known as nunataks. Only two flowering plant species grow here—Antarctic hair grass and Antarctic pearlwort—joined by around 100 moss species, 25 liverwort species, and 300–400 lichen species. Due to the region's extreme climate, where icy winds blow year round and average annual temperatures range from 14°F to -76°F (-10°C to -60°C), some algae and lichen species survive inside rocks, growing within tiny pores.

▽ FLORAL CUSHION

Antarctic pearlwort grows in cushionlike clumps, reaching a height of around 2 in (5 cm).



The McMurdo Dry Valleys

One of Earth's most extreme desert areas, where organisms survive severe cold, arid conditions



The McMurdo Dry Valleys lie west of the McMurdo Sound, bordering the Ross Sea. Covering just 0.03 percent of the continental landmass, they contain the largest ice-free region in Antarctica and constitute an extreme cold desert ecosystem, where temperatures can plunge to -90°F (-68°C) and winds rip through at up to 200 mph (322 kph). The surrounding mountains block the valleys from the East Antarctic Ice Sheet, and humidity levels are extremely low, yet microbial life survives here in all but the driest areas.

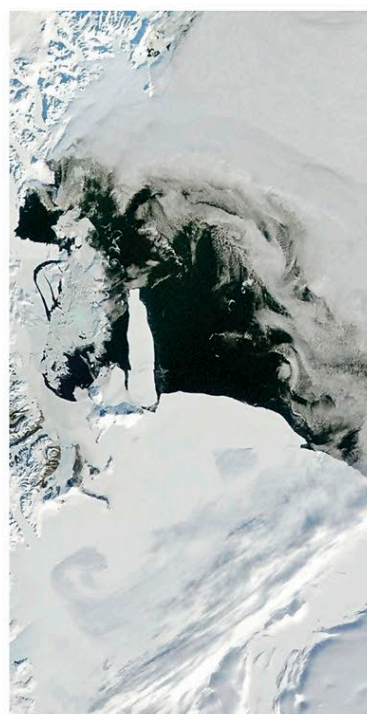
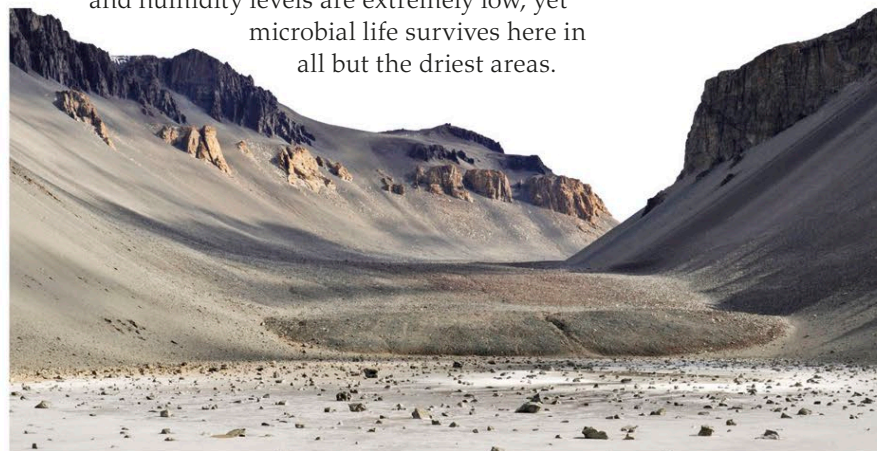


△ STAINED LANDSCAPE

The outflow of a saline lake, the Blood Falls, gets its red color from the water's high iron content.

◁ SUPER-SALTY

The shallow Don Juan Pond is so salty it does not freeze even in winter, unlike other water bodies in the dry valleys.



△ BIGGEST ICEBERG

In 2000, the largest iceberg ever recorded, called B-15 (to left of center), broke off the Ross Ice Shelf (at bottom in this image).